4996804826642777801

Hi, how's it going? I just received your email. I think so, yeah. Yeah, well, my comments.

Yeah, thank you. So basically, you have the new file. These are the changes that you made.

Did you find the link file? Yeah, I uploaded it. If you see it down in the email, you will be able to download it. Yeah, I did.

It's too big. But I call it D5. But there's some corrections in there, I think, on a file.

But the thing is, just some main comments. So I think paper is great. So I had just a little paragraph of description about like the GSR.

So although we're assigning to Jennifer paper, it's the base of which we are doing. Yeah, it's like we present the sensor itself and just referring to a paper. I asked you to.

There's one point because I don't have the paper citation and all this stuff. But I think like if you look at the end, there's a please add here. Yeah, it's basically the same thing.

But for the part where you classify sign language here. Oh, yeah, we don't. Yeah, you type the paper.

There is no explanation. So try to do just like a little summary of that. You know, mostly based on that.

And if you don't understand it. Yeah. Yeah.

OK, yeah, you're right. I, you know, was getting too long. I sort of skipped that.

And the thing is that I don't like 87 different objects. They're mostly corrected. But like classified correctly.

So. No, no, no, no. Yeah, it's OK.

But still, it's an important part. People need to get the description. And you could cut to somewhere else in the paper.

And that's one of the things. What is the maximum length? You could add for the maximum length of the paper. I don't think there is no limitation here.

No, I think I'm right. Yeah. And I prepared like a one column version.

And it's almost 30 pages. Because they said for the initial submission. You can do it like a single page.

And it's easier for them to review. Yeah. 30, I think 30 pages.

One of the things, like, I could. So I have access to. I don't know if you heard about.

No. Super powerful model. It was so powerful that they restricted the access.

They restricted the access. And gave access to some company to batch. Because you could find.

You could find a lot of issue with websites. Like critical path that you could exploit. And stuff like that.

So you get access to some company. So that you can batch your stuff. So now they are starting to open this.

It's super expensive to run. And only accessible through. What is this called? Just read about it.

Close. M-Y-E-H-O-S. So the point is.

It's an AI like you probably never tried. I could ask him to compress a paper of my liking. Called the two page.

I think it would probably be better. Because there's some parts that are very long in the explanation. I think cutting to the point.

But I would not. So I could have him work through the latex file. And do it himself.

So if you do this. You could aim for like ten pages. And look at the results.

And I think it's going to be better than you and me. Cutting what is important and what is not. Okay.

Yeah, sure. You know, I can read the journal web page and see if there's limitations. I couldn't find any.

Because usually just ask on the internet. Google will answer you. It's RASC.

And I think RASC is ten or 12 max. Or it could be. Six to 12 pages.

Yeah. It's what I thought. It's 12 is the max.

And we are. This is. We're 13.

But people at the max, people don't love it. So I will try to compress it to 11. And then I will send you a D6 instead of D5.

Okay. Before me addressing this. You already fixed that, right? No.

I already did the whole thing that I told you in the file. But the last point. So I didn't have the necessary information to do it.

Right. So I only need to address this last one. And these are all.

No, but not. By the way, the uncan still see. Yeah.

It's really like the. Like. Just a little paragraph that explains that part.

Yeah. I can. Okay.

That would be great. If you can. Update my D5 that I did.

Just with a paragraph on this. Sure. And then I will ask the effective model to compress it.

Or could you do the compression and send it back to me? But just leave a little bit of space for me to maybe add a small confusion matrix or just one. Okay. I will do the opposite.

Actually send me a paragraph describing this. What I have. Okay.

And then I will ask him to introduce it at the right place in the text and then compress the whole thing. Okay. Okay.

I'll do that. Yeah. I'll do that by tomorrow.

Yeah. And it's not going to miss out anything or. No.

No. No. No.

No. No. No.

No. No. No.

No. No. No.

No. No. Just to run METAS from a new document, it's going to cost like $100, but it's okay, I have access.

On cell 22 June, I have access with the contraband that can help them, but it's just going to be crazy. If you go look on the web, people are like programming, but it's going to be very sensitive to run. We evaluated internally that to give access to people that code software to that model would cost us about $8,000 per month per engineer.

That's why they were trying to get access on cell 22. If you look, it's crazy. People basically build themselves video games, ultra nice video games.

We've built it in an iPhone app. You could basically give all your paper at the end of your thesis and say, generate me a good PowerPoint for a thesis defense. It will just do it for test, like the best you can imagine.

It's just, it is access. It does open browser, go search on the web, download data, can come back. It will do all the tools.

There's pictures you update, it will update the picture. By the way, in the comments, be careful in the next paper. One of the things in that paper, usually in academic papers, you need to have a neutral thumb.

For example, if you have an extraordinary improvement, it's usually, you should say, non-negligible, which is more like saying, it's more like academic. Extraordinary is subjective, it's saying it's very awesome. Non-negligible is to say, if you look at the stats, it's not just a small difference.

That's more like, try to be more… It's not our job to work up… Well, complimented, but not expected. I mean, you can do the job by saying, like I said, non-negligible, but not. It's supposed to be neutral.

It's a scientific approach. It's like, you're not doing marketing, you're doing… You're just reporting. So, you report.

Yeah, I had the same comment from Jean-Philippe for the first paper. I tried to implement it here, but I think there are still… Well, I changed it now. I changed it.

Oh, perfect. That's… Okay, I'm going to add just one paragraph. Just send it to me.

Don't add it. Just make it a tiny paragraph on that, and I will introduce it. I mean, I'm not going to put the confusion matrix.

I'm just going to say that this is like… Almost 98, 99% of our test points are classified correctly, and that's why I… So, we don't have the result of what if it's classified wrong. We don't have that, because most of them are classified correctly. Otherwise, I wouldn't start the whole search thing.

No, no, but you can also explain that. Sure. Oh, great.

So, that's for the paper, and then you will be able to submit. What about… My kid, you talked about what you would like to address next. Sure.

Or something we talked today, or… Yeah. First of all, I talked with Beric yesterday. Yep.

And he said that he uploaded the simulator for new sensors. Yep. But not the old sensor, but he's going to do the old sensors as well.

So, that we're going to have an elaborate new sensor for… And, yeah, I was thinking maybe there are a couple of things that we can do. When we have the simulation, we can do the whole, like, large-scale, lots of data. We're re-grasping validation, the thing that we have.

So, we can just replace the extrapolation with the simulation, and maybe report the results again. And also, I was thinking maybe we can add another factor to the optimizer, which is the task-oriented graph. So, we have the more stable graph, maximizing GSR.

Also, we're minimizing the efficiency. And also, we are maximizing, this is the third term that we're adding to the cost function, which is task-oriented objective. And it could be based on the shape.

Could we just… Okay. What is the thing at this moment for the re-imagination? We only validated with the GSR. It's like, yeah, we extrapolate this, and that is supposed to come back.

And we went very plain with three different shapes. And to do with… …we have vision, but we don't do… …simulation data… …because in simulation, we have the objective… …we can take an object and then change the position everywhere around. Like, you graph, you take the position, and you take the data all around.

And then you have your initial, and what is the reality for other tactile. And then, you could do that for a lot of object, and have something learning of what is the right imagination when you detect a given contact. I don't know if you follow me.

Yeah, we can do that. So, you graph, and I think partially in simulation, because it's easy to do. In real life, it would imply that the object don't move, and then you go graph 10 times around your point.

So, basically, we'd have the original contact, and what is the real thing around. And instead of mapping to three different shapes of object and imagining, we could have a complex CNN or a complex network that would map those outputs. So, at the end, we have a model that can get the initial contact, and output the extended contact.

Yeah, a bit like what you're doing at this moment in the simulator, because we tried to build a deterministic model of how the sensor compressed in the relation capacitance, but at the end, we were not able to build a deterministic model. So, basically, it took input, output, and then used the CNN to learn it. It reproduced the output signal from an input pretty good.

In your case, it would be two. And what I would do is probably having, at this moment, we'll always work with only one snapshot. And the thing is, I think there's a lot to learn into the shape of the object by taking, like, four or five snapshots as enclosed.

Like, as soon as you detect, like, a fraction of a newton, take a snapshot, you take one until you reach full stop. But in between, there's different, like, you would see the formation pattern. And I'm saying this because we did a paper with Jean-Philippe, who was my student, and Jennifer, another student, one that will actually be a new professor in the electrical department next year.

But one of the things we realized is to recognize that we're using objects, like, with similar, to recognize objects in itself, taking three or four snapshots, I could send you the paper, was the one, was the modality that was giving the most, the better output. Yeah. So anyway, if we could train a classifier that would be, because at this moment, what we do is basically to go map different objects and then to contrapolate.

And maybe that's still the right approach. Maybe we would have, like, a better mapping to the object, because right now sometimes it's hard to distinguish the flat, like, a cuboid to a giant cylindrical feature. But I guess if you take different snapshots, you would see the difference, because you would see that it's really start, and the way that, like, the contact evolves is very different.

So I think that could be, and then simulation would be easy, because you could run, like, set up, like, hundreds of objects, and then we learn on it. That's right. And try to see if it's mapped in reality.

But that could be an approach. But we do the whole training simulation, and we get the model out, and we can verify it with the... Yeah, yeah, yeah. But the output of the whole thing is just to have a new model that can predict beyond.

But, yeah, but for me, like, maybe I'm confused, but the idea of the whole thing is that we don't have the CAD model, and we don't have vision, we don't have the CAD model. No, no, no. This is the whole novelty of the whole thing.

Yeah, and if you work in Isaacson with this, you don't have the vision, and that's the point. To do that, we probably have these visions. It's like, we just go and grab.

We just go learn the truth. If we move, what is the actual... But we have the 3D model of the object. We have the STL file.

Yeah, but we just use it inside of the, to generate the contact point on... Yeah, we use the STL file for simulation only to do the training. And... Yeah, so... So, here's the cylinder, that if you take it, you generate different images until you stop. You rest it.

But then, in a virtual world, you could go and re-rest it in all directions. Just like, randomness. We just shift by one millimeter in every direction.

It will generate, like, 100 different contacts around the initial. So, let's say that is the initial, and then we move one meter this way, one meter this way, one, one... We always do the same thing. Obviously, if you move around that axis, the cylinder is going to generate the same thing.

But the thing is, we want the network to learn that. When you see a super straight contact like that, it will map instantly to... Just to understand, instead of... Because right now, you need to map to an object, and then you need to create, like, what should be the contact around that. Let's let the AI create this extended... So, basically, the output of the network input would be this, and the output would be, like, this sensor four times bigger.

Like, all the part around... Basically, what we do with the... Yeah, geometry... But we train using sims. But it doesn't contradict the first paper. I mean, somebody can say... Yes and no.

It's like, the first one, we had a deterministic approach, and then we try a full... And the thing is, the first one we didn't access, like, doing this on a real robot would have been very complicated. And also, you have always, like, the uncertainty around the sensor itself. Now, you have a constant behavior that includes noise that replicates the sensor, but you have that behavior, and then it's easy to generate tens of thousands of data to give to your CNN and learn.

But we had this access. That's the first digital twin of the force of tactile sensor. I guess if we had that at the beginning, we would have went this way.

We would have had a simulator with tactile sensor. So, I couldn't see yourself do, like, tens of thousands, like, or 100,000 data force. So, I don't think at all it contradicts.

And it maybe just builds something more robust, where, like, at the end, it creates very, very accurate representation of the object. And the contribution is just introducing this model and verifying it at the end. Yeah, well, it's like the grasping model.

And I see it very similar. It's before, like, and Jen is not the only one to have used machine learning, but with the GSR you use, there's plenty of people that build a deterministic model to evaluate if it's stable or not. Right.

And one of the thing is we saw that it's highly correlated with the center of mass. That's right. And the area of contact.

So, we could have used that as a deterministic model. Actually, one of the first thing we started to make as a model with one of my first students was deterministic using those feature. And so, the point is, it's kind of what we did with most deterministic.

We used AI to map, to create the fine shape. And then, we're extrapolating. But now, if we have the possibility to generate 10,000 to 100,000 of data in simulation, because it's super fast generating all those things and super accurate in a way that's, like, even one millimeter, that's very good.

So, it could be actually, like, it's kind of like if Jen, on his first paper, did, like, a deterministic model and then beat it with the AI model. But it also, what she did somehow, she started with a very simple CNN model for GSR and then increased the amount of data and the diversity of shape of the data she took. And then, by the way, if we do that, I would use the shape of the object that she used for this.

Oh, you mean the… Yeah. That is the… That is your object, right? Those are, yeah, but these are… This was Jen's. Yeah.

Oh, sorry. That's okay. I guess I can take one or two.

So, but that's part of the thing. We could also wait. The thing is, maybe it's a discussion we should have had with Jean-Philippe because he's only going to come back in July and I don't want you to lose a month.

So… But still, I think I can spend some time learning about the simulation and how to integrate it with my code. The only thing is, yeah, there's some issue right now, but I think right now, personally, if we… The issue is it's bad at generating friction, so it's hard to lift objects, but here we will not lift objects. So that's one of the reasons I thought about that experiment because, like, we consider the object fixed, like here, and we just go around and look at it.

So, did you understand my idea? Yeah. Like, we really go, we grasp, we start from a point, we grasp, and then we go take all the search maps, the search grids. Like, I think for the first paper, you found it that GSR is good on that, like, that zone around the initial contact.

So let's say it's a 2 millimeter or, like, not 2 millimeters, but let's say 1 centimeter and a half on each side around the initial contact. Let's say that. So we go take all those different patterns, and we work not with only one image.

We work with, like, three images. A series of… Yeah, it's like… Yeah. Yeah, it's really to capture… It's really to capture, like, it seems like the pattern is evolving as we compress.

Yeah. It goes quick. I should be able to… Is it on robotics, you think? The video is… No, the video is not.

I don't know if John has generated it, but the GitHub is linked to… Yeah, I have the gif for the Coral Lab. If that's available there. By the way, the GPMO… It's like fire.

It was trained in the night to recognize the object. Are these new objects in robots? No, no, no. We train it on them.

It's recognized, but some of them generate pretty similar patterns. Okay. Like a square, when you take it this way, it looks like a… Could be the same as the… That's not the very best, but there's a tiny difference, like, or it could be the star.

Sometimes star we're confused with, depending on how you compare it, but still, like, the star was more sharp than the angle, than the square. So, by the way, that was one. This one, I did a code for the angle, which is using the value.

I improved it along the way. Yeah, yeah. I was thinking about something like this, like a task-oriented optimization.

So, try to grasp it with maximum stability and also the readiness for the… These are the new stuff. Yeah. These are really better, I think.

Yeah, yeah, yeah. And the… Yeah. There is a system to get the noise when you're reading.

Right. This is really… This is something, like, with first sensor, I would make a perfect example. That was pretty tough, just for humans to be able to do that.

Right. Like, the last one is pretty tough. I was not able as a human.